

Héctor Hernández Cruz

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SUMMARY

Senior AI / Cloud Engineer with 5+ years designing and deploying production AI systems on cloud-native infrastructure across healthcare, SaaS, and ecommerce domains. Deep expertise across both stacks: AI/LLM engineering (LangChain, LlamaIndex, LangGraph, RAG pipelines, OpenAI API, AWS Bedrock, GCP Vertex AI, pgvector, RAGAS) for intelligent applications, and cloud-native infrastructure (AWS, Azure, GCP, Kubernetes, Terraform) for scalable and cost-optimized AI workload deployment. Experienced designing LLMOps pipelines covering model versioning, evaluation harnesses, and drift monitoring. Full-stack fluency across Python, Node.js, and React/TypeScript enables end-to-end AI product ownership. Consistent measurable impact: 42% API latency reduction in AI-augmented clinical systems and zero-downtime AI deployments in a HIPAA-regulated environment.

WORK EXPERIENCE

Curai Health

Senior Full-Stack Engineer

Jul 2024 – Present

San Francisco, CA, United States

- Designed and deployed the full AI infrastructure for a HIPAA-regulated clinical platform, architecting multi-stage RAG pipelines with LangChain and LlamaIndex on AWS (ECS, Lambda, S3, RDS pgvector), multi-model LLM orchestration across AWS Bedrock (Claude, Titan), GCP Vertex AI, and OpenAI GPT-4o with fallback routing and cost-optimized model selection, achieving a 42% reduction in AI API latency through Redis semantic caching and async Python (FastAPI) inference.
- Established LLMOps practices including MLflow model versioning, LangSmith A/B testing for RAG quality, RAGAS automated retrieval evaluation, embedding drift monitoring, and LangGraph agentic workflows with human-in-the-loop approval gates for HIPAA-regulated clinical decision support.
- Deployed AI services on AWS EKS and Azure AKS via Terraform, Helm, ArgoCD, and GitHub Actions CI/CD, implemented IAM, KMS, and GuardDuty for HIPAA-compliant AI data handling, and contributed Node.js and React/TypeScript features for the patient intake platform.

Ashby

Software Engineer

Jun 2022 – Jun 2024

San Francisco, CA, United States

- Built Python and Node.js backend services for the core ATS product, integrating pgvector for candidate data search and Redis semantic caching with PostgreSQL query optimization reducing API response times by 36%, and developed async pipelines for high-volume candidate data workflows.
- Maintained AWS EKS and Azure AKS multi-cloud infrastructure via Kubernetes, Terraform, and GitHub Actions CI/CD, and expanded pytest and Cypress coverage across critical ATS service boundaries.

Bazaarvoice

Software Engineer

Jul 2019 – Apr 2022

Austin, TX, United States

- Optimized data processing pipelines on AWS (EC2, ECS, S3, RDS) for high-traffic ecommerce features, reducing data sync latency by 11%, and reengineered PostgreSQL analytics queries cutting report generation from 8 seconds to 1.2 seconds (85% improvement).
- Managed AWS production infrastructure with Terraform IaC and GitHub Actions CI/CD, and led the full codebase migration from JavaScript to TypeScript establishing type-safe service contracts.

Self-Employed Contractor

Freelance Software Engineer

Jan 2017 – May 2019

Austin, TX, United States

- Designed and delivered AI-augmented cloud applications for startups and SMBs across the US, integrating NLP and ML APIs into Python and Node.js services on AWS Lambda, ECS, and S3.
- Built React frontend interfaces for retail and hospitality clients, delivering production-ready cloud-native systems on AWS (EC2, RDS, S3).

EDUCATION

The University of Texas at Austin

Bachelor of Science, Computer Science

Aug 2015 – May 2019

Austin, TX, United States

KEY SKILLS

- AI / LLM Stack:** LangChain, LlamaIndex, LangGraph, LangSmith, OpenAI API (GPT-4o, Embeddings), AWS Bedrock, GCP Vertex AI, Azure OpenAI, Semantic Kernel, Guardrails AI, PEFT / LoRA
- RAG & Vector Search:** pgvector, Pinecone, Weaviate, Qdrant, Chroma, FAISS, Hybrid Search (BM25 + Dense), Reranking, RAGAS Evaluation, Embedding Drift Monitoring
- LLMOps & MLOps:** MLflow, Weights & Biases, LangSmith, SageMaker, Vertex AI Pipelines, Apache Airflow, Great Expectations, Streaming (SSE, WebSockets)
- Cloud Platforms:** AWS (ECS, EKS, Lambda, RDS, S3, SQS, Bedrock, IAM, KMS, GuardDuty, CloudFront), Azure (AKS, OpenAI Service, Azure ML), GCP (Vertex AI, GKE, BigQuery ML)
- Infrastructure & DevOps:** Kubernetes, Helm, ArgoCD, Terraform, Docker, GitHub Actions, GitOps, Prometheus, Grafana, Datadog, OpenTelemetry, FinOps
- Backend & APIs:** Python (FastAPI, asyncio, Pydantic), Node.js (NestJS, LangChain.js), Go, REST APIs, GraphQL, gRPC, PostgreSQL (pgvector), Redis, Apache Kafka
- Frontend & Security:** React, TypeScript, Next.js, Vercel AI SDK, HIPAA Compliance, OWASP LLM Top 10, PII Redaction, Responsible AI, mTLS, RBAC